

AllPath Trading Agent — Technical One-Pager

github.com/dukesky/allpath-trading-agent · MIT · 2,368 tests · self-hosted: FastAPI + SQLite + curated-markdown memory, no external services beyond LLM + Alpaca

Options Alpha Agents
Alpaca AI Trading Agents
Hackathon

AI LOGIC — THE LLM PLANS; CODE TRADES

Three model tiers, one agent. `CHAT_MODEL` (claude-sonnet-5) does conversation, research, and strategy drafting; `REVIEW_MODEL` (haiku) analyzes soft-rule triggers; `MEMORY_MODEL` (claude-opus-5) runs nightly reflection and memory consolidation — the highest-stakes writes get the strongest model.

Strategies are the interface between model and machine. The agent writes YAML: a prose *thesis* plus deterministic *rules* — conditions compiled by a whitelist-AST evaluator (no `eval`) over `{price, position_qty, position_weight, avg_entry_price, pnl_pct}`; actions in a closed grammar (`buy $3000 · sell all · buy_call $1500 dte>=5 otm=3% · close_options`). A scheduled sentinel evaluates every armed rule each 30 min at **zero LLM cost**; triggers are one-shot. This week every rule was **hard** — **the intraday trading path contained no LLM calls at all**. The inference budget went to thinking: a bounded nightly reflection session (12 tool calls, wall-clock deadline) reviews the day against each thesis, writes lessons into four layers of curated markdown memory, re-arms spent triggers, and proposes revisions as diffs through one guarded pipeline — byte-exact base-staleness check, strict version increase, and a freeze barring reflection from touching `authorization/status`: **the agent cannot promote itself to more autonomy**. Over 5 days: **12 self-revisions auto-applied** — night one it found a stop→re-entry loop in all five of its own strategies; mid-week it invented salvage rules that recovered **~\$1,165** of decaying premium; its final report critiqued its own stop placement.

RISK GATES — DETERMINISTIC, NON-BYPASSABLE

`Executor.execute()` / `execute_option()` are the **only** paths to a broker; the LLM can only produce intents. Every intent passes a plain-code gate: $\text{per-order} \leq \$5,000$ · $\text{position weight} \leq 25\%$ · **options exposure $\leq 10\%$ of equity** · cash-reserve floor · daily trade cap. Asymmetry by design: sells and closes are exempt from value caps — **risk reduction is never blocked**. Above the per-order gate: a **drawdown circuit breaker** (peak-tracked equity, -15% trips once → all `auto` strategies demote to confirm, alerts everywhere, exits still execute) · a **DTE ≤ 1 expiry sweep** (no option rides into exercise) · **market-hours discipline** (closed-market option triggers stay armed instead of burning into venue rejections) · **honest failure** (a broker response without an order id journals an `error` and pages the operator — never a phantom fill; this guard caught a real pre-market rejection on day one). Authoring-time validation closes the loop: option actions require `auto` + hard rules, and option entries must carry paired `close_options` exits.

ALPACA INFRASTRUCTURE

Stocks: `alpaca-py` `TradingClient` (paper) with socket timeouts bound onto its session; notional market orders, immediate fill-poll, full journaling. **Options:** **100% through Alpaca's official MCP server** (`uvx alpaca-mcp-server`, `stdio`) as a managed subprocess — lazy spawn, per-spawn ownership handles, auto-respawn-once, 30 s call timeouts, clean teardown. Contract selection is deterministic code, not model whim:

```
rule fires → get_option_contracts(expiry ≥ today+dte, right) → nearest expiry → strike ≈ spot×(1±otm) →
get_option_latest_quote → qty = [budget/(ask×100)] → RiskGate.check_option →
place_option_order(buy_to_open | sell_to_close, market, day) → journal → notify
```

Every trade, rejection, skip, LLM decision, and nightly report lands in a local SQLite journal — **the audit trail is the demo**.

Week scoreboard (honest): fresh \$100k paper account · one kickoff chat, then five days hands-off · 17 rule-driven fills, 0 human trading decisions · -4.9% (\$95,138): stock book ≈ flat ($-\$180$); the 5-DTE options overlay paid $-\$4.7k$ tuition — a flaw the agent diagnosed in writing on night one and spent the rest of the week correcting itself.